Complex Numbers

ClipSAT Printable Practice Suite (clipsat.org) • Topic 9.01

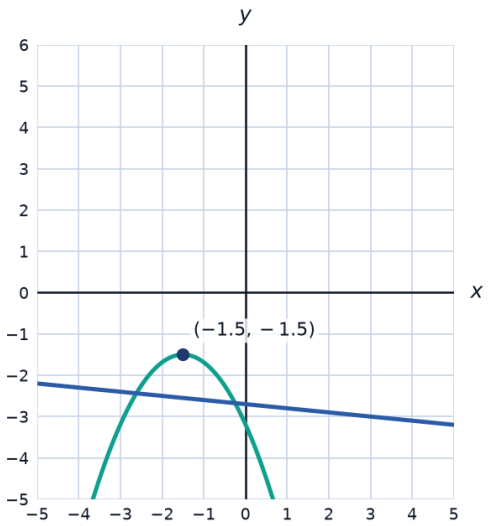
Part I — Free-Response Questions (FRQs): Foundational Fluency (1–7)

1 **[Foundational] • Target: The complex conjugate and division**

Find the conjugate of $z = 7 - 2i$, and compute $z\bar{z}$.

2 **[Foundational] • Target: The complex conjugate and division**

Simplify $\frac{3+i}{2-3i}$ by multiplying by the conjugate of the denominator.



3 **[Foundational] • Target: Modulus and the Argand diagram**

Find the modulus $|z|$ of $z = 5 - 12i$.

4 **[Foundational] • Target: Modulus and the Argand diagram**

The graph shows several complex numbers plotted on the Argand diagram (real part horizontal, imaginary part vertical).

a Read off the complex number plotted at $(4, 1)$ in Cartesian form.

b Find the modulus of the complex number plotted at $(-2, -3)$.

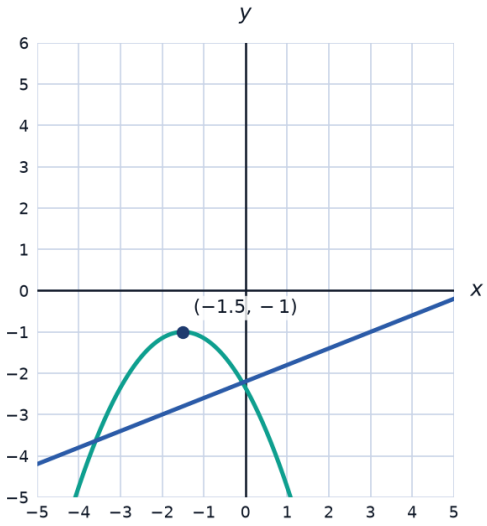
5 **[Foundational] • Target: Solving equations with complex roots**

Solve $x^2 + 2x + 5 = 0$, giving your answer in the form $a + bi$.

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6 [Foundational] • Target: Solving equations with complex roots

One root of a quadratic equation with real coefficients is $3 - 4i$. State the other root, and find the quadratic equation.



7 [Foundational] • Target: Solving equations with complex roots

Given that $z = 1 + 2i$ is a root of $z^3 - 4z^2 + 9z - 10 = 0$ (with real coefficients), find the other two roots.

Part II — Free-Response Questions (FRQs): Intermediate Multi-Step Application (8–14)

8 [Intermediate] • Target: Converting to polar (modulus-argument) form

Write $z = \sqrt{3} + i$ in polar form $r(\cos\theta + i\sin\theta)$, giving the argument in radians.

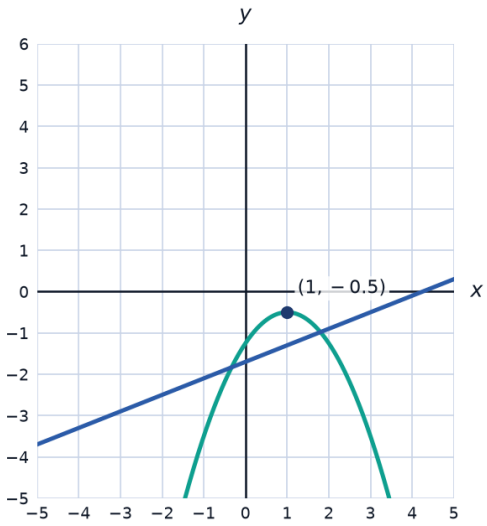
9 [Intermediate] • Target: Converting to polar (modulus-argument) form

Convert $z = 6\left(\cos\frac{2\pi}{3} + i\sin\frac{2\pi}{3}\right)$ to Cartesian form.

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10 [Intermediate] • Target: Multiplying and dividing in polar form

Given $z_1 = 4\left(\cos\frac{\pi}{3} + i\sin\frac{\pi}{3}\right)$ and $z_2 = 2\left(\cos\frac{\pi}{12} + i\sin\frac{\pi}{12}\right)$, find $z_1 z_2$ and $\frac{z_1}{z_2}$ in polar form.



11 [Intermediate] • Target: Multiplying and dividing in polar form

Find the modulus and argument of $z = \frac{(1+i)^3}{1-i}$, using De Moivre's theorem for the numerator and polar division.

12 [Intermediate] • Target: De Moivre's theorem

Use De Moivre's theorem to find $\left[3\left(\cos\frac{\pi}{6} + i\sin\frac{\pi}{6}\right)\right]^4$, giving your answer in Cartesian form.

13 [Intermediate] • Target: Finding roots using De Moivre's theorem

Find the three cube roots of $z = 27\left(\cos\frac{\pi}{2} + i\sin\frac{\pi}{2}\right)$, giving arguments in $[0, 2\pi)$.

14 [Intermediate] • Target: Finding roots using De Moivre's theorem

Find all four fourth roots of -16 (i.e. solve $z^4 = -16$), giving arguments in $[0, 2\pi)$.

Part III — Free-Response Questions (FRQs): Challenging & Cross-Lesson Synthesis (15–20)

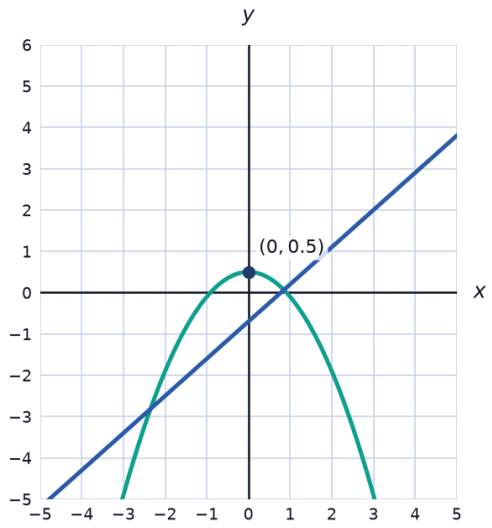
15 [Challenging] • Target: A well-designed word problem: AC circuit impedance • Cross-Lesson: 1.01 Functions: Modulus & Inverse

This question has three parts. An AC circuit has impedance $Z = R + iX_L$, where $R = 8\,\Omega$ (resistance) and $X_L = 6\,\Omega$ (inductive reactance). (a) Write Z in Cartesian form, find $|Z|$, and find $\arg Z$ to the nearest degree. (b) The current is $I = \frac{V}{Z}$, where $V = 100$ volts (real). Find I in the form $a + bi$, by rationalizing the denominator. (c) Find $|I|$, to two decimal places.

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16 [Challenging] • Target: De Moivre's theorem • Cross-Lesson: 10.01 Numerical Methods

Convert the Cartesian equation $(x^2 + y^2)^2 = 4(x^2 - y^2)$ into polar form and identify the name and key features of the resulting curve.



17 [Challenging] • Target: Finding roots using De Moivre's theorem • Cross-Lesson: 11.01 Differential Equations

[Polar Coordinates, Complex Numbers & De Moivre's Theorem · Practice 3] If the governing relation for this system is $g(t) = 5t + 5$, evaluate $g(8)$ and find the inverse input t when $g(t) = 50$.

18 [Challenging] • Target: A well-designed word problem: AC circuit impedance • Cross-Lesson: 12.01 Poisson Distribution

Find all five solutions to $z^5 = -32$ in polar form $re^{i\theta}$ with $-\pi < \theta \le \pi$. Show that the sum of these five roots equals 0.

19 [Challenging] • Target: Basic arithmetic with complex numbers • Cross-Lesson: 13.01 Continuous Random Variables

Level 2 · MediumSolve the transcendental equation analytically for x in exact form using natural logarithms: $3^{2x+1} = 5^{x-2}$.

20 [Challenging] • Target: The complex conjugate and division • Cross-Lesson: 14.01 Sampling & Estimation

Express $\frac{3+\sqrt{5}}{2-\sqrt{5}}$ in the form $p + q\sqrt{5}$, where p and q are rational numbers to be found.

Part IV — Multiple-Choice Questions (MCQs): Foundational Concept Mastery (21–28)

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21 [Foundational] • Target: Basic arithmetic with complex numbers

A complex number z is represented in polar form as $z = r(\cos(\theta) + i\sin(\theta))$. If a complex number has a modulus $r = 5$ and an argument $\theta = \frac{\pi}{3}$, what is its rectangular form $a + bi$?

a. $\frac{5}{2} + i\frac{5\sqrt{3}}{2}$

b. $5 + i5$

c. $\frac{5\sqrt{3}}{2} + i\frac{5}{2}$

d. $5i$

22 [Foundational] • Target: The complex conjugate and division

A complex number z is represented in polar form as $z = r(\cos\theta + i\sin\theta)$. Given that $z^3 = 8i$, and z lies in the first quadrant such that its argument θ satisfies $0 < \theta < \frac{\pi}{2}$. Find the value of $r^2\cos\theta$.

a. 2

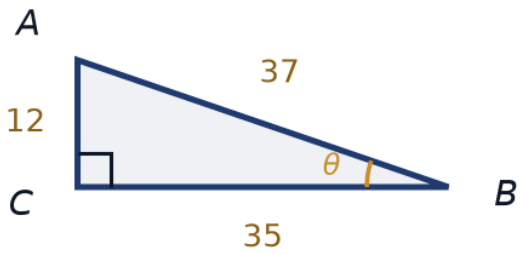
b. 4

c. $2\sqrt{2}$

d. 1

23 [Foundational] • Target: Modulus and the Argand diagram

Let the complex number z be represented in polar form as $z = r(\cos\theta + i\sin\theta)$. Suppose z satisfies the equation $z^3 = -64i$. If z is restricted to the fourth quadrant (i.e., $\frac{3\pi}{2} < \arg(z) < 2\pi$), find the real part of z.



a. 4

b. -4

c. $2\sqrt{3}$

d. $-2\sqrt{3}$

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24 [Foundational] • Target: Solving equations with complex roots

Let the complex number z_1 be represented in polar form as $z_1 = r_1(\cos\theta_1 + i\sin\theta_1)$, where $r_1 = 4$ and $\theta_1 = \frac{\pi}{3}$. Let another complex number, z_2, be represented as $z_2 = 2(\cos\theta_2 + i\sin\theta_2)$. If $z_1 \cdot z_2 = 8(\cos\alpha + i\sin\alpha)$ and $\alpha = \frac{5\pi}{6}$, find the value of θ_2 in radians, given that $0 \le \theta_2 < 2\pi$.

a. $\frac{\pi}{2}$

b. $\frac{2\pi}{3}$

c. $\frac{5\pi}{6}$

d. $\frac{\pi}{6}$

25 [Foundational] • Target: Converting to polar (modulus-argument) form

Consider the complex number $z = \left(\frac{\sqrt{3}}{2} + \frac{1}{2}i\right)^{10}$. If z is expressed in polar form as $r(\cos\theta + i\sin\theta)$, where $r > 0$ and $0 \le \theta < 2\pi$, find the exact values of r and θ.

a. $r = 1, \theta = \frac{5\pi}{3}$

b. $r = 1, \theta = \frac{4\pi}{3}$

c. $r = 10, \theta = \frac{5\pi}{3}$

d. $r = \sqrt{3}, \theta = \frac{4\pi}{3}$

26 [Foundational] • Target: Multiplying and dividing in polar form

Consider the complex number $z = r(\cos\theta + i\sin\theta)$. If $z^3 = -8$ and $r > 0$, and the principal argument of z is $\frac{\pi}{3}$, find the value of r.

a. 2

b. 4

c. 8

d. 16

27 [Foundational] • Target: De Moivre's theorem

A complex number is represented by the point $P(3, 4)$ in the complex plane. What is the modulus of this complex number?

a. 5

b. 7

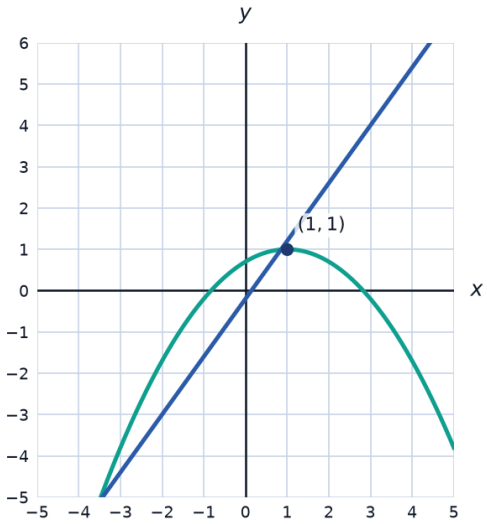
c. 25

d. 16

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28 [Foundational] • Target: Finding roots using De Moivre's theorem

How many distinct complex nth roots does the equation $z^5 = 1$ have, and what is the angle (in degrees) between consecutive roots on the Argand diagram?



a. 5 roots, 72° apart

b. 4 roots, 72° apart

c. 5 roots, 60° apart

d. 6 roots, 72° apart

Part V — Multiple-Choice Questions (MCQs): Intermediate Exam Application (29–37)

29 [Intermediate] • Target: A well-designed word problem: AC circuit impedance

Consider the complex number $w = 4 - 3i$. Find the modulus of w, denoted as $|w|$.

a. 5

b. 6

c. 7

d. 25

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30 [Intermediate] • Target: Basic arithmetic with complex numbers

Let $z = x + yi$ be a complex number. The modulus of z, denoted by $|z|$, is given by $|z| = \sqrt{x^2 + y^2}$. Given that $|z| = \sqrt{5}$ and the real part of z is 2, find the possible values for the imaginary part of z.

a. $\pm\sqrt{3}$

b. ± 3

c. $\pm\sqrt{1}$

d. ± 1

31 [Intermediate] • Target: The complex conjugate and division

Consider the complex number $z = 3 + 4i$. What is the modulus of z?

a. 5

b. 7

c. 25

d. $\sqrt{7}$

32 [Intermediate] • Target: Modulus and the Argand diagram

Consider the complex number $z = 3 + ki$, where k is a real parameter. The modulus of z is given by $|z| = \sqrt{13}$. Find the positive value of k.

a. 1

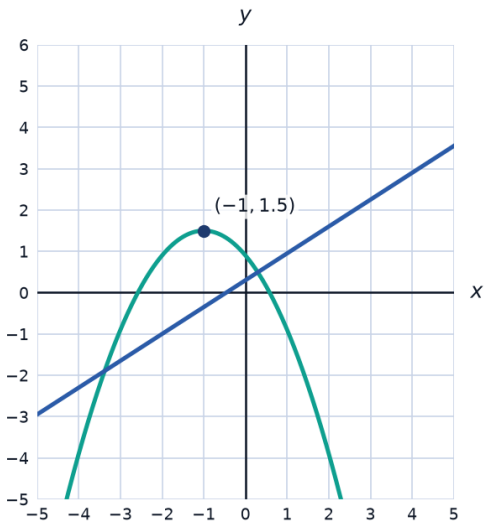
b. 2

c. 3

d. 4

33 [Intermediate] • Target: Solving equations with complex roots

Consider the complex numbers $z_1 = 5 - 2i$ and $z_2 = 1 + 4i$. Find the value of $z_1 - z_2$ in the form $x + yi$.



a. 4 - 6i

b. 6 + 6i

c. 4 + 6i

d. 6 - 6i

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34 **[Intermediate]** • **Target: Converting to polar (modulus-argument) form**

Consider the complex number $z = -3 + 4i$. Find the modulus of z, denoted by $|z|$.

a. 5

b. 7

c. 25

d. 16

35 **[Intermediate]** • **Target: Multiplying and dividing in polar form**

Using De Moivre's theorem, evaluate $(1 + i)^9$.

a. $16 - 16i$

b. $16 + 16i$

c. $-16 - 16i$

d. $-16 + 16i$

36 **[Intermediate]** • **Target: De Moivre's theorem**

Product of roots of $2x^2 - 5x + 3 = 0$:

a. $\frac{3}{2}$

b. $\frac{5}{2}$

c. $-\frac{3}{2}$

d. 3

37 **[Intermediate]** • **Target: Finding roots using De Moivre's theorem**

Find all cube roots of unity.

a. $1,\ \frac{-1+i\sqrt{3}}{2},\ \frac{-1-i\sqrt{3}}{2}$

b. $1, i, -1$

c. $1, -1, i$

d. $1, i\sqrt{3}, -i\sqrt{3}$

Part VI — Multiple-Choice Questions (MCQs): Challenging & Cross-Lesson Synthesis (38–45)

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38 [Challenging] • Target: A well-designed word problem: AC circuit impedance • Cross-Lesson: 12.01 Poisson Distribution

For $x^2 - x - 56 = 0$, find the product of the roots.

a. -42

b. -47

c. -56

d. -51

39 [Challenging] • Target: Basic arithmetic with complex numbers • Cross-Lesson: 13.01 Continuous Random Variables

Simplify $(1 + 2i)(5 + 5i)$, giving your answer in the form $x + yi$.

a. $-3 + 16i$

b. $-5 + 15i$

c. $-1 + 12i$

d. $-3 + 14i$

40 [Challenging] • Target: The complex conjugate and division • Cross-Lesson: 14.01 Sampling & Estimation

For $x^3 - 19x - 30 = 0$, find the product of the three roots.

a. 30

b. 43

c. 27

d. 20

41 [Challenging] • Target: Modulus and the Argand diagram • Cross-Lesson: 15.01 Hypothesis Testing

Write $z = 4(\cos 180^\circ + i\sin 180^\circ)$ in the form $x + yi$ (exact values).

a. $-4 + 1$

b. 4

c. -4

d. $-4i$

42 [Challenging] • Target: Solving equations with complex roots • Cross-Lesson: 2.01 Exponentials & Logarithms

Parametric area using Green's theorem: $A = \frac{1}{2}\oint(x\,dy - y\,dx)$. For ellipse $x = a\cos t, y = b\sin t$:

a. $\pi ab/2$

b. πab

c. $2\pi ab$

d. ab

43 [Challenging] • Target: Converting to polar (modulus-argument) form • Cross-Lesson: 3.01 Advanced Trigonometry

Find $\bar{z}$ (the complex conjugate) if $z = -3 + 9i$.

a. $-3 - 9i$

b. $-3 + 9i$

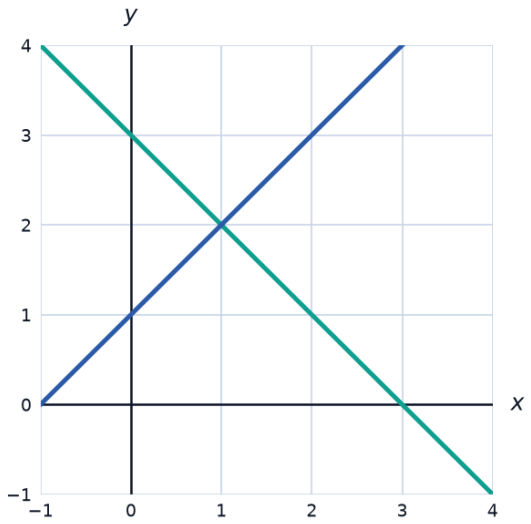
c. $3 - 9i$

d. $3 + 9i$

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44 [Challenging] • Target: Multiplying and dividing in polar form • Cross-Lesson: 4.01 Differentiation

The diagram shows the graphs of two linear equations. The point of intersection of the two lines represents the solution to the system of simultaneous equations. Find the coordinates of the point of intersection.



a. (1, 2)

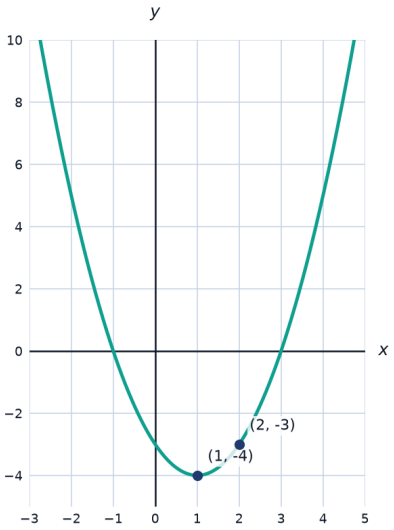
b. (2, 1)

c. (3, 0)

d. (0, 3)

45 [Challenging] • Target: De Moivre's theorem • Cross-Lesson: 5.01 Integration

The diagram shows the graph of the function $y = f(x)$. The function is of the form $f(x) = x^2 + bx + c$. The graph passes through the point $(2, -3)$ and has a minimum value of -4 at $x = 1$. Find the value of b.



a. -2

b. 1

c. 2

d. -1

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